

LRU Page Replacement Algorithm

Completed

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing

Select Language : C (GCC 9.2.0)

```

1 #include<stdio.h>
2 int main(){
3     int n,frames;
4     printf("Enter number of pages in reference string:");
5     scanf("%d",&n);
6     int ref[n];
7     printf("Enter the reference string: \n");
8     for(int i=0;i<n;i++){
9         scanf("%d",&ref[i]);
10    }
11    printf("Enter number of frames: \n");
12    scanf("%d",&frames);
13    int frame[frames];
14    int time[frames];
15    for(int i=0;i<frames;i++){
16        frame[i]=-1;
17        time[i]=0;
18    }
19    int faults=0;
20    int hits=0;
21    int c=0;
22    for(int i=0;i<n;i++){
23        int found=0;
24        for(int j=0;j<frames;j++){
25            if(frame[j]==ref[i]){
26                hits++;
27                c++;
28                time[j]=c;
29            }
30            found=1;
31        }
32        if(found==0){
33            faults++;
34            // Replace the least recently used page
35            int min_time = time[0];
36            int min_index = 0;
37            for(int k=1;k<frames;k++){
38                if(time[k]<min_time){
39                    min_time = time[k];
40                    min_index = k;
41                }
42            }
43            frame[min_index] = ref[i];
44            time[min_index] = c;
45        }
46    }
47    printf("Total Page Faults: %d\n", faults);
48    printf("Total Page Hits: %d\n", hits);
49    return 0;
50 }
```

Test Cases

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2